ZAFIR MARICAR

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Skills

- **Languages:** C++, Python, Java, Kotlin, Lua, TypeScript
- **AI & Video:** NVIDIA CUDA, Huawei Ascend (CANN), GStreamer, FFmpeg, RTSP, OpenAI / Anthropic / Gemini APIs
- **Backend & Cloud:** Docker, RabbitMQ, Redis, MariaDB, Node.js, Firebase, Azure, CI/CD, Kubernetes
- **Graphics & Simulation:** OpenGL, NVIDIA PhysX, ImGui, FMOD, AnyLogic
- **Web & Mobile:** Next.js, Tailwind CSS, React, Android Studio

Experience

AI Engineer — **NCS** — **2025 – Present**

- Architected GStreamer/FFmpeg pipelines for RTSP stream and file ingestion, powering real-time AI inference.
- Built Docker microservices with RabbitMQ, Redis, MariaDB and Python for scalable video chunking.
- Reduced video processing latency by 30% through multi-threading and memory buffer optimizations.
- Optimized parallel AI inference workloads across NVIDIA CUDA and Huawei Ascend (CANN) architectures.
- Deployed end-to-end AI video analytics products on-site and demoed live applications at industry conferences.

Research Engineer Intern — **Hitachi Asia** — **2024 – 2025**

- Developed proof-of-concept simulations in AnyLogic (Java-based) in collaboration with the mobility team.
- Modelled MRT station interchanges, integrating human behavior, pedestrian flow and transport infrastructure.
- Co-authored and presented an award winning research paper based on the metro simulation I created at a forum.

Police Officer, Ground Response Force (National Service) — **Singapore Police Force** — **2019 – 2021**

- Responded to time-critical emergencies; independently patrolled and manned neighborhood police posts.
- Took part in covert law enforcement operations and served in the COVID-19 special task force.

Education

Master of Technology (Part-time, Scholarship) — **National University of Singapore** — **2025 – Present**

- Major in Software Engineering | **CGPA: 5.0 / 5.0**

Bachelor of Science with Honours — **Singapore Institute of Technology** — **2021 – 2025**

- Major in Computer Science in Real-Time Interactive Simulation (with DigiPen Institute of Technology)

Diploma — **Temasek Polytechnic** — **2016 – 2019**

- Major in Aerospace Electronics

Projects

Torque Engine | 3D Game Engine

- Led as Product Manager, directing 7 programmers and 3 designers to build a 3D game engine from scratch.
- Built in C++, Lua and Python using OpenGL, ImGui, FMOD, NVIDIA PhysX and Azure.
- Solely responsible for implementing the engine's physics and networking systems.
- Used the engine to develop a physics-intensive kart racing game in the style of Mario Kart.

Foodkaki | AI Web App, AngelHack × DBS Hackathon

- Led a team of 4 to build a generative AI web app that guides users' financial decisions on food purchases.
- Developed with Next.js, Node.js, Tailwind CSS, Python and the Gemini API.
- Praised for functionality: integrated the FairPrice API to pull live product prices and data into the app.

IPPT Buddy | Android Fitness App

- Built a real-time tracker for sit-ups, push-ups and runs in Android Studio using Kotlin and Firebase.
- Owned sensor programming, using the gyroscope, accelerometer and proximity sensors.

Publications

Mo, Z., **Zafir, M.**, & Bahoy, G. L. (2025). "Influence of Selective Security Check on Heterogeneous Passengers at Metro Stations." *Engineering Proceedings.* https://doi.org/10.3390/engproc2025102003

- **3rd Place, Best Paper Award** – ITS Asia-Pacific Forum 2025, Suwon, South Korea.